Weekly Progress: Embedding & RAG

On Development of an on-Premise AI Product Recommendation Model Using Open-Source LLM

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Project Description

Project Title

Development of an on-Premise AI Product Recommendation Model Using Open-Source LLM

Company: Narosu Co. Ltd.

Project Description

- The e-commerce market is becoming increasingly competitive, requiring advanced recommendation systems.
- Large marketplaces use sophisticated AI models, but independent online store operators lack such solutions.
- This project aims to develop an **AI-powered product recommendation system** tailored for self-managed online stores.
- By analyzing customer preferences, the system will suggest products that align with their interests.

Project Objective

- Develop an AI-based search and recommendation system that analyzes user input sentences and recommends the **TOP 5** most relevant products based on semantic similarity.
- Utilize Natural Language Processing (NLP) and vector search technology to provide a highly refined search experience.
- Deploy in an on-premise environment to ensure data privacy and full control over the recommendation process.

Company Requirements

핵심 기술 스택

컴포넌트	사용 기술	설명
LLM (대형 언어 모델)	Llama 3.1 8B	문맥을 기반으로 자연어 이해 및 생성
LangChain (LLM 프레임워크)	LangChain	LLM과 외부 데이터 소스를 연결하여 AI 응답의 문맥을 강화
임베딩 모델	OpenAI text-embedding-ada-002	의미론적 유사도를 기반으로 텍스트 임베딩 생성
벡터 DB	무관	고차원 벡터 데이터 저장
벡터 검색 라이브러리	FAISS	대규모 벡터 데이터 검색 최적화
RAG (Retrieval-Augmented Generation)	RAG 기반 검색 + LLM	실시간 검색을 통해 문서를 참조하여 정확한 답변 생성
API 아키텍처	REST API (FastAPI or Flask or Django)	클라이언트-서버 간 데이터 통신을 위한 API 구축
GUI (그래픽 사용자 인터페이스)	챗봇	사용자가 보기 편한 직관적인 챗봇 템플릿 적용

- OpenAI text-embedding-ada-002 is a paid model

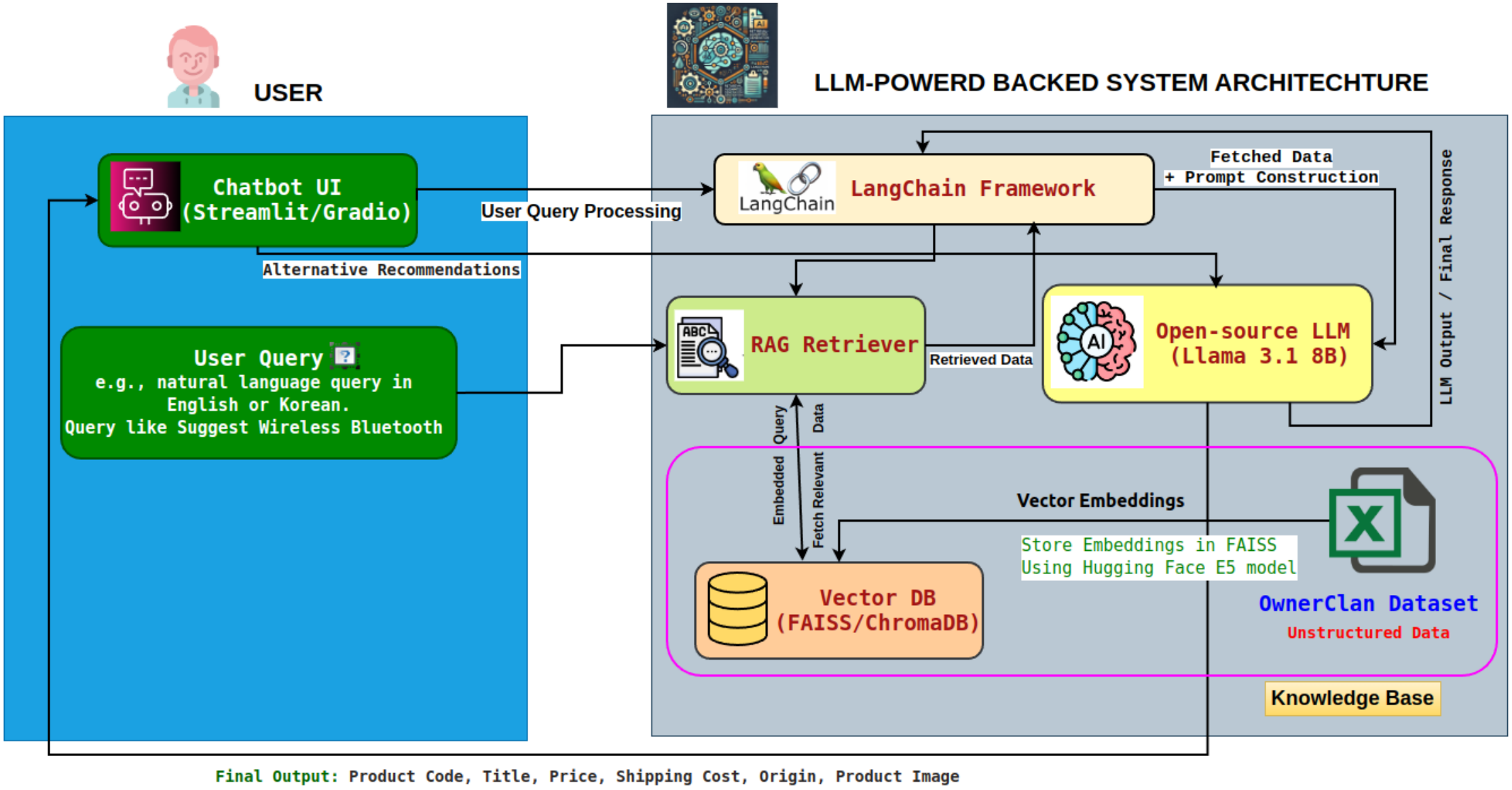
Try with free models such as:

- Sentence Transformers (`all-MiniLM-L6-v2`)
- Hugging Face Transformers (`intfloat/e5-small-v2`)

Project Schedule (Gantt Chart)

Phase	2025													
	March				April				May				June	
	1	2	3	4	1	2	3	4	1	2	3	4	1	2
Project Understanding and Data Collection														
Data Analysis and Pre-processing														
AI Model Selection and Training														
System Development														
Chatbot UI Development														
Testing and Deployment														

System Configuration (AI Model) Diagram



Communication with company

Questions:

- Does the recommendation system need to support multilingual responses (Korean, English, Japanese), etc.? Or is there a language you prefer?
- What are the key metrics to evaluate the performance of a recommendation model?
- If a customer rejects the first set of recommendations, does the Chatbot have to offer an alternate recommendation?

Contents of the answer

- The dataset is fixed in Korean, but it must be multilingual. Responses must be submitted in a language supported by your Llama model (Priority: Korean, English)
- F1 Score 를 제외한: If there is a noteworthy evaluation index in AI evaluation similar to product recommendation, please present it. I would appreciate it if you could write a brief explanation of the indicator and the reason why you selected the indicator.
- Alternative recommendation is required . Promptly, send your students their thoughts on what questions they would ask to find information about their preferred products. (vector update questions, vector orientation questions, vector adjustment questions, etc.)

Communication with company (Cont.)

Questions:

- Does the recommender system need to support multilingual responses (English, Japanese, Korean), etc.? Or is there a language you prefer?
- Are datasets updated regularly? If so, how often will new products be added?

Contents of the answer

- The dataset is fixed in Korean, but it must be multilingual. When entering a sentence in Korean, the recommendation and answer must be given in Korean, and if the sentence input is a foreign language such as English or Thai , the answer must be submitted in the appropriate language of each language.
- In the AI convergence project, the secondary dataset is final. but, Our company's ultimate goal is to update products regularly like tie commerce . Therefore, by subdividing the goals, the first goal is 4 million data, and the second and final goals are 4 million data considering the updating(updating) of the product. We would appreciate it if you could share your ideas on how to update and update the RAG-based chatbot easily, quickly, and conveniently.

Communication with company (Cont.)

Questions:

- What are the key metrics to evaluate the performance of a recommendation model?
- If a customer rejects the first set of recommendations, does the chatbot have to offer an alternate recommendation?
- Can I use a paid embedding model (OpenAI's GPT)?

Contents of the answer

- Based on F1 Score. Based on the precision and recall of the AI recommendation system, we evaluate whether the AI recommendation is practical.
- Alternative recommendation is required. The goal is to allow you to proceed to checkout with only the chatbot's recommendation without looking at the product list, so if you reject the first set of recommendations, you need to ask additional questions or suggest recommendations in a different direction.
- You can change to OpenAI's paid embedding model. (Please note that Narosu does not support subsidies.)

Communication with company

There is additional sharing content, so we will write it together .

- When sharing the secondary dataset , the contents of the column list of the existing Excel file will be organized and shared. Please proceed with the chatbot development based on the Excel file.
- https://revf.tistory.com/303 (AI 오픈소스 정리본).
- We will send you a well-organized link, so we will organize the frameworks that students will use and tell us.

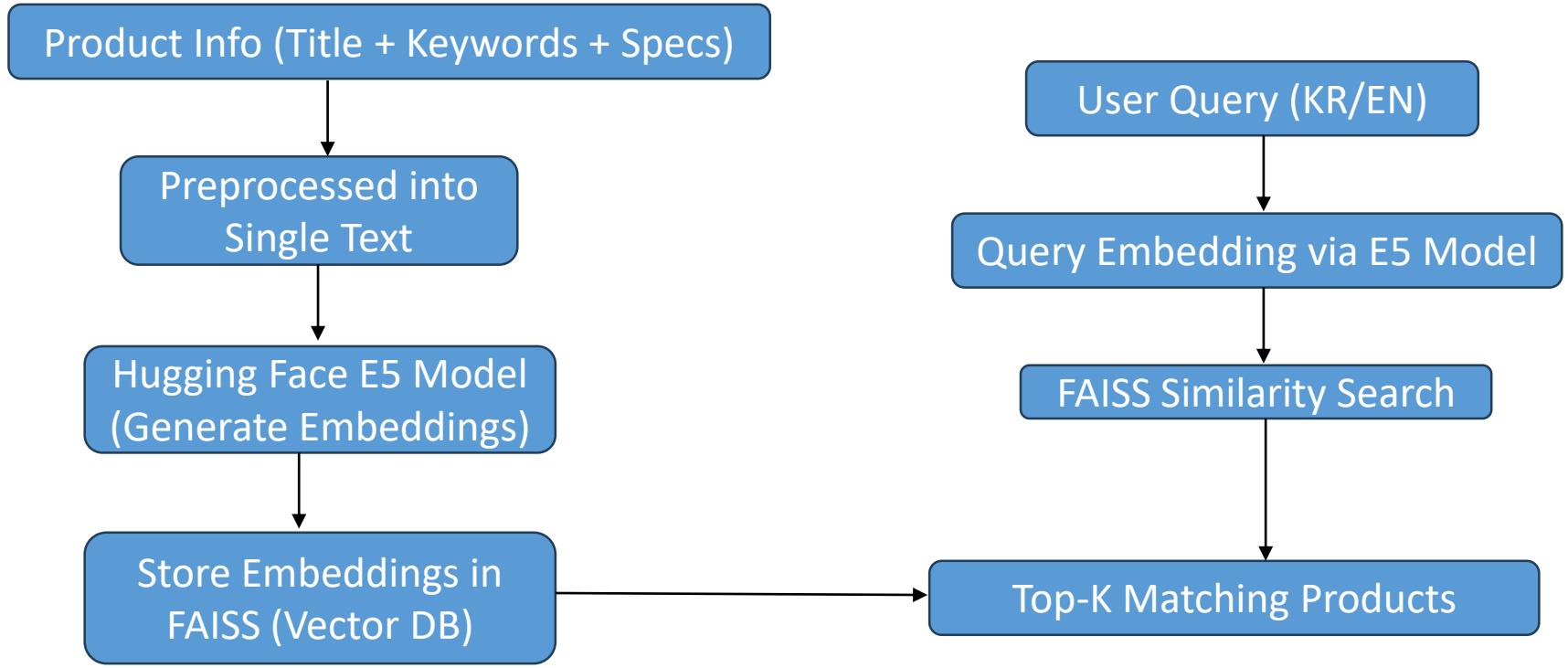
Embeddings in our system (OwnerClan Dataset)

🔍 What are Embeddings?

- Embeddings are **numerical representations** of text data.
- Similar meanings produce **similar vector representations**.
- Example: "블루투스 이어폰" and "Bluetooth earphone" → close in vector space.

🎁 How Embeddings Work

- **Input**: Product title + keywords + specs → combined text
- **Model**: Free E5 model generates vector embeddings
- **Storage**: Vectors stored in FAISS (vector DB)
- **Search**: User query → embedded → matched with product vectors



Need to usability check of FAISS in my project!

Example of Embeddings Output in our system

Original Query (KR): 게이밍 블루투스 헤드셋

Translated Query (EN): Gaming Bluetooth Headset

원본상품명 Product Name (EN)	키워드 Keywords (EN)	카테고리명 Category (EN)
6953 맥스틸 올인원 게이밍 헤드셋 GM-GHC300 헤드셋, Max Steel All-in-One Gaming Headset GM-GHC300 Headset	블루투스헤드셋,무선헤드셋,유선헤드셋,게이밍헤드셋 Bluetooth headset, wireless headset	led헤드셋,올인원헤드셋,삼... 디지털/가전>멀티미디어장비>PC헤드셋 Digital/Home Appliances> Multimedia Equipment

Problem of embedding model, "all-MiniLM-L6-v2" does not support multiple languages.

```
Original Query (KR): 갤럭시 스마트폰
Translated Query (EN): Galaxy smartphone

                원본상품명                                                  키워드  ...                                                Keywords (EN)                                                Catego
ry (EN)
6953  맥스틸 올인원 게이밍 헤드셋 GM-GHC300  헤드셋,블루투스헤드셋,무선헤드셋,유선헤드셋,게이밍헤드셋,led헤드셋,올인원헤드셋,삼...  ...  Headset, Bluetooth headset, wireless headset, ...  Di
gital/Home Appliances> Multimedia Equipment>...
```

Query output from FAISS sometimes provides incorrect results."

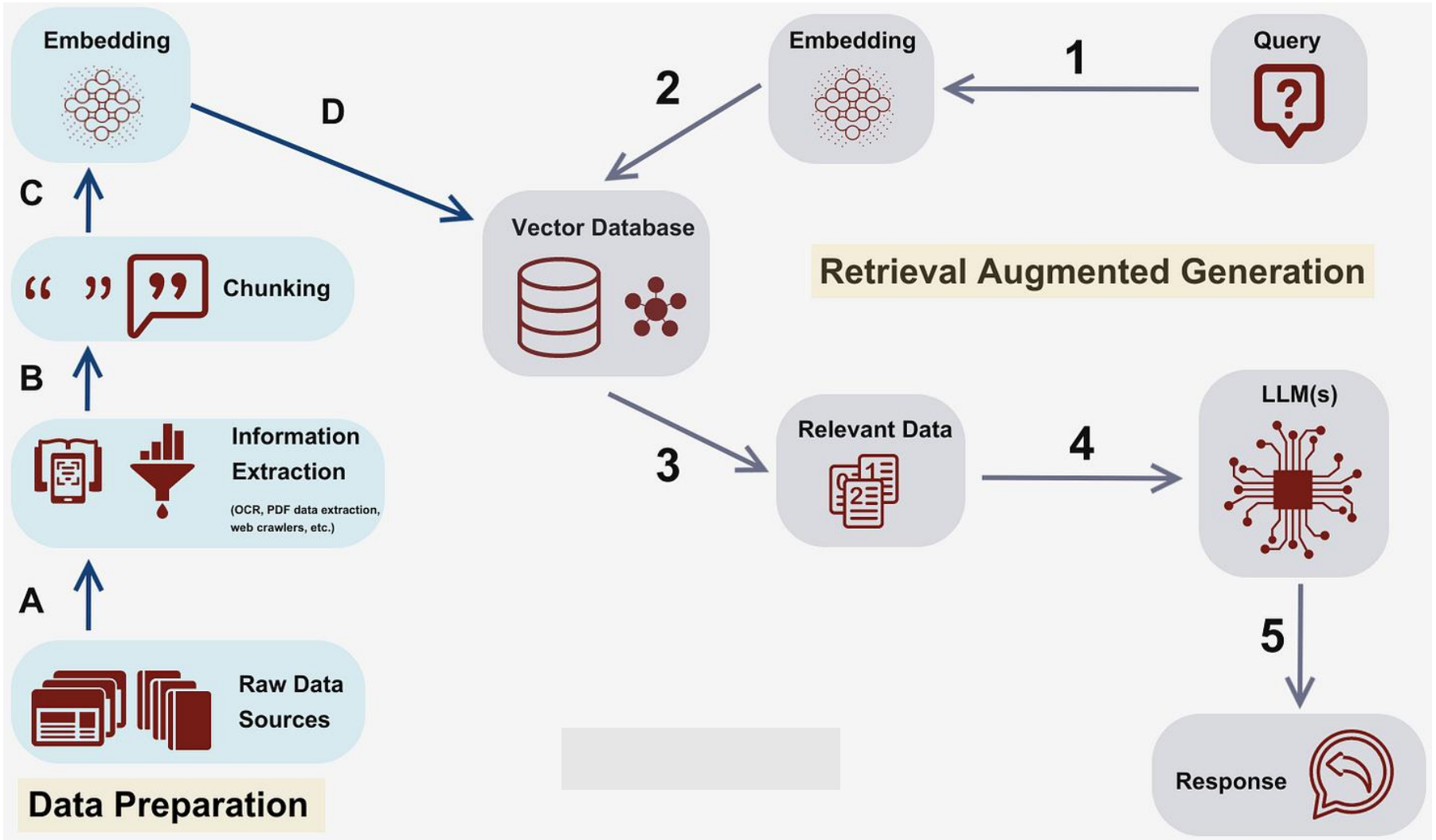
Retrieval-Augmented Generation (RAG) Study

What is RAG?

- Combines **retrieval** and **generation**.
- RAG retrieves relevant product info and passes it to an LLM (like LLaMA 3) to generate **context-aware responses**.
- Useful when dealing with large product databases or vague queries.

RAG Pipeline in this Project

- **User Query**: Given in English or Korean (e.g., "남자용 데일리 백팩 추천").
- **Embedding**: The query is embedded using the same E5 model.
- **Retrieval (FAISS)**: Top-k most similar product entries are retrieved.
- **LLM + LangChain**: Retrieved data is sent to LLaMA 3 with a crafted prompt.
- **Final Output**: A personalized product suggestion or alternative recommendation.



Next Tasks To Do

Next Step

- Vector Database model selection for usability and scalability.
- Text Embedding model selection for multiple languages such as Korean, English, and Chinese.
- Test the models based on the OwnerClan dataset.

Thank You for Listening